Data Warehouse (DWH) & Characteristics:

A **Data Warehouse (DWH)** is a central repository of integrated data from multiple sources, specifically structured for query, reporting, and analysis.

Characteristics:

1. **Subject-Oriented** – Data is organized around major subjects like sales, customers, products, etc., rather than daily operations.
2. **Integrated** – Data from different sources is combined, standardized, and stored in a consistent format to provide a unified view.
3. **Non-Volatile** – Once data is entered, it is not updated or deleted. Only new data is added, which maintains historical records.
4. **Time-Variant** – Each piece of data is time-stamped, enabling trend analysis and historical comparisons over time.
5. **Analysis-Oriented** – Designed mainly for queries, decision-making, and business intelligence rather than transaction processing.

Dimensional Modelling:

Dimensional modelling is a design technique used in DWH to make data simple and efficient for analysis.

1. **Fact Table:**
   - **Stores Quantitative Data** – Contains numerical measures like sales amount, profit, or revenue that represent business performance.
   - **Foreign Keys + Large Size** – Includes foreign keys linking to dimension tables; usually very large in size as it stores transactional/aggregated data.
2. **Dimension Table:**
   - **Stores Descriptive Data** – Contains attributes like product name, customer name, location, and time that provide context to facts.
   - **Smaller & Detailed** – Usually smaller than fact tables and stores descriptive details for better analysis and reporting.
3. **Simple and Understandable** – Data is structured in a way that business users and analysts can easily understand and query.
4. **Relationships** – Fact tables are linked with dimension tables using keys, enabling powerful analysis of data.

Star Schema:

The **Star Schema** is the simplest and most commonly used schema in dimensional modelling.

1. **Central Fact Table** – Stores numerical measures (sales, revenue, etc.).
2. **Dimension Tables** – Surround the fact table and describe facts (time, product, customer, region).
3. **Single-Join Queries** – Queries need only one join between fact and dimension, making them faster.
4. **Simplicity** – Easy to design, understand, and use for reporting.
5. **High Performance** – Denormalized dimensions and fewer joins make it efficient for analytical queries.
6. **De-normalized Dimensions** – Dimension tables store all descriptive attributes in one table, avoiding further splitting.

Snowflake Schema:

The **Snowflake Schema** is a more complex form of schema where dimension tables are normalized.

1. **Central Fact Table** – Same as star schema, holds business measures.
2. **Primary Dimension Tables** – Connected directly to the fact table.
3. **Secondary Dimensions** – Dimension tables may be split further (e.g., product → product & product category).
4. **Reduced Redundancy** – Normalization avoids duplicate data and saves storage.
5. **Complex Queries** – More joins are required compared to star schema, making queries slower.

OLAP vs OLTP

Aspect	OLAP (Online Analytical Processing)	OLTP (Online Transaction Processing)
1. Purpose	Used for analysis, reporting, and decision support. Helps managers analyze trends and make strategic decisions.	Used for day-to-day operations like inserting, updating, and deleting transactions in real-time.
2. Type of Data	Works on historical, consolidated data stored in a data warehouse. Mostly read-only.	Works on current, real-time data in operational databases. Frequently updated.
3. Database Design	Uses de-normalized schema (Star/Snowflake) to reduce joins and improve query performance.	Uses normalized schema (3NF) to reduce redundancy and maintain data integrity.
4. Queries	Handles complex queries with aggregations, joins, and multidimensional analysis. Example: "Yearly sales growth by region."	Handles simple, short queries for transactions. Example: "Update account balance after withdrawal."
5. Users & Usage	Used by business analysts and managers for reporting, forecasting, and strategic decisions.	Used by clerks, cashiers, and operational staff for daily business processes.

OLAP Operations:

1. **Drill Down** – Move from summary to detailed view (Year → Quarter → Month → Day).
2. **Roll Up** – Opposite of drill down; aggregate data for higher-level summaries (Day → Month → Year).
3. **Slice** – Fix one dimension and analyze a sub-cube (e.g., sales only for 2024).
4. **Dice** – Select sub-cube with conditions on multiple dimensions (e.g., Electronics in Asia in 2024).
5. **Pivot/Rotate** – Reorient the cube to view data differently (e.g., Sales by Region → Sales by Product).

Data Mining & KDD:

Data Mining – The process of extracting hidden patterns, relationships, and trends from large datasets using algorithms and statistical techniques.

KDD (Knowledge Discovery in Databases) – The complete process of extracting useful knowledge from data.

Steps in KDD:

1. **Selection** – Identify and extract relevant data from large databases (e.g., customer sales data).
2. **Preprocessing** – Clean and prepare data, handle missing values, remove noise, and correct inconsistencies.
3. **Transformation** – Convert data into suitable format (e.g., normalization, aggregation, feature engineering).
4. **Data Mining** – Apply algorithms like classification, clustering, association, and regression to find patterns.
5. **Evaluation** – Validate discovered patterns for accuracy and usefulness.
6. **Interpretation** – Present results in user-friendly formats (charts, reports, dashboards).
7. **Iteration** – Process may repeat until meaningful knowledge is discovered.
8. **Actionable Insights** – Final results must support better decision-making.
9. **Integration** – Knowledge is integrated into business processes for operational use.
10. **Applications** – Used in fraud detection, marketing, healthcare, finance, recommendation systems, etc.

